



TEAM: Crisis Swarm

Member Details

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Theme

Theme 2 – Security in the Agentic Future

PROBLEM STATEMENT



The Human-Dependent Crisis (5-Step Manual Cycle)



The Staggering Cost & Core Gap



⚡ **The Core Gap:** There is no system today that autonomously detects, investigates, diagnoses, remediates, and secures cloud incidents end-to-end — without human intervention.



SOLUTION

Introducing CrisisSwarm — The Autonomous AI Incident Response Swarm





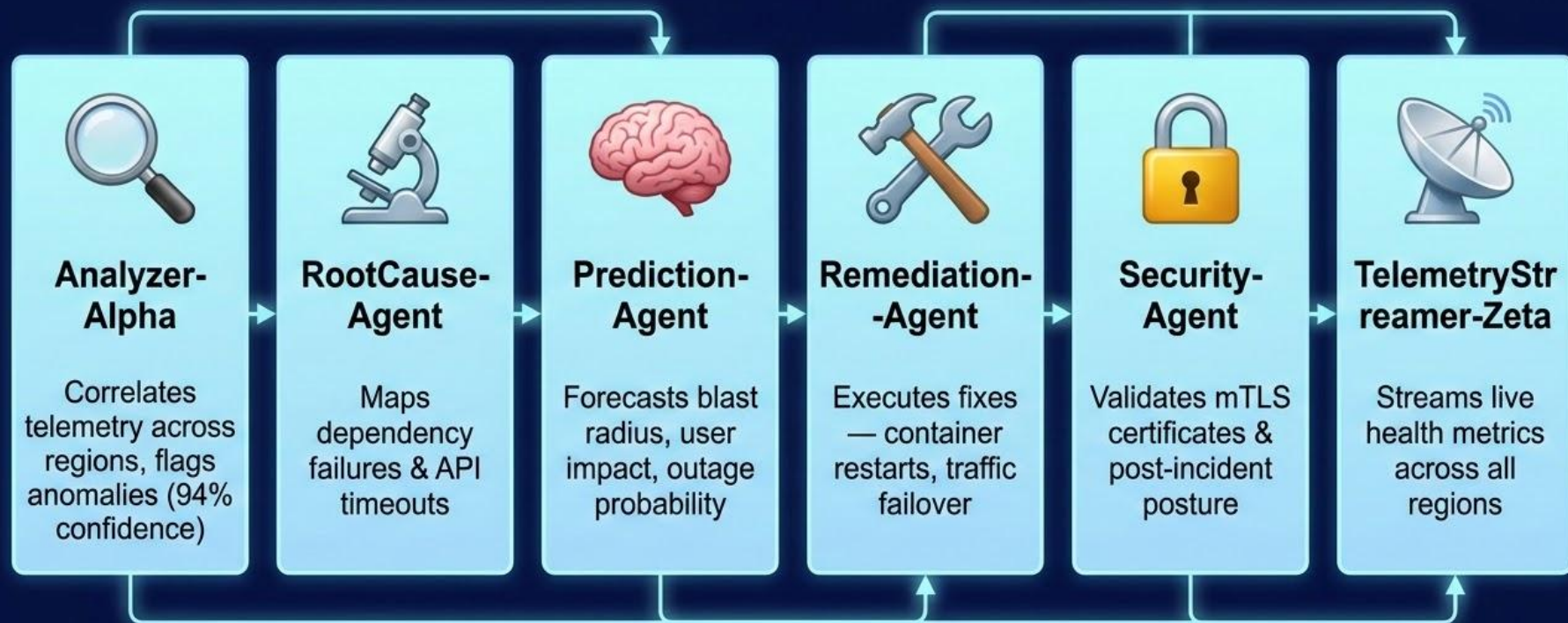
SOLUTION



SOLUTION



The Six-Agent Swarm





SOLUTION

Differentiators



Zero-Human Autonomy —
no manual steps.



Operator Controls —
global stop/resume +
per-agent toggles.



Azure OpenAI Powered —
domain-specific prompts
keep agents reliable.



Real-Time Dashboard —
global stop/resume + with live SSE logs.

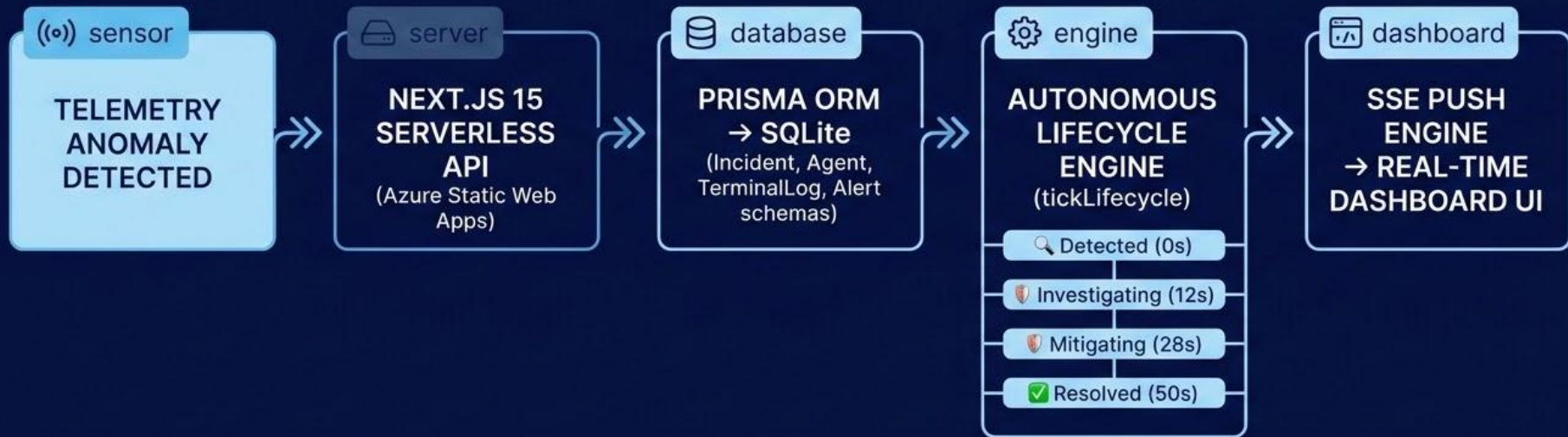


Azure OpenAI Conterid —
glassmorphic UI prompts even
without credentials.



Fallback Simulation Engine — demo-ready even without credentials.

METHODOLOGY ARCHITECTURE





METHODOLOGY TECHNOLOGY STACK



Frontend

Next.js 15 (React), TailwindCSS, Glassmorphic UI



Backend API

Next.js Serverless API Routes



Database

Prisma ORM + SQLite



AI Engine

Azure OpenAI `gpt-4o` with specialized domain prompts



Real-Time Streaming

Server-Sent Events (SSE)



Hosting

Azure Static Web Apps



Testing

Vitest (100% pass rate — 6 unit & integration tests)

METHODOLOGY DESIGN DECISION & CONSTRAINT



Why 6 Agents, not 1?

Specialized, isolated agents with strict domain prompts solve for 10,000+ log lines, Kubernetes rollback scripts, and auditing simultaneously, preventing context failures and hallucinations.



Why SSE instead of WebSockets?

Server-Sent Events are simpler, stateless-friendly for serverless deployments on Azure, and optimal for one-directional real-time log streaming.



Why SQLite + Prisma?

Lightweight, zero-config, fully embedded persistence that scales for the demo and can be migrated to Azure SQL or Cosmos DB for production.

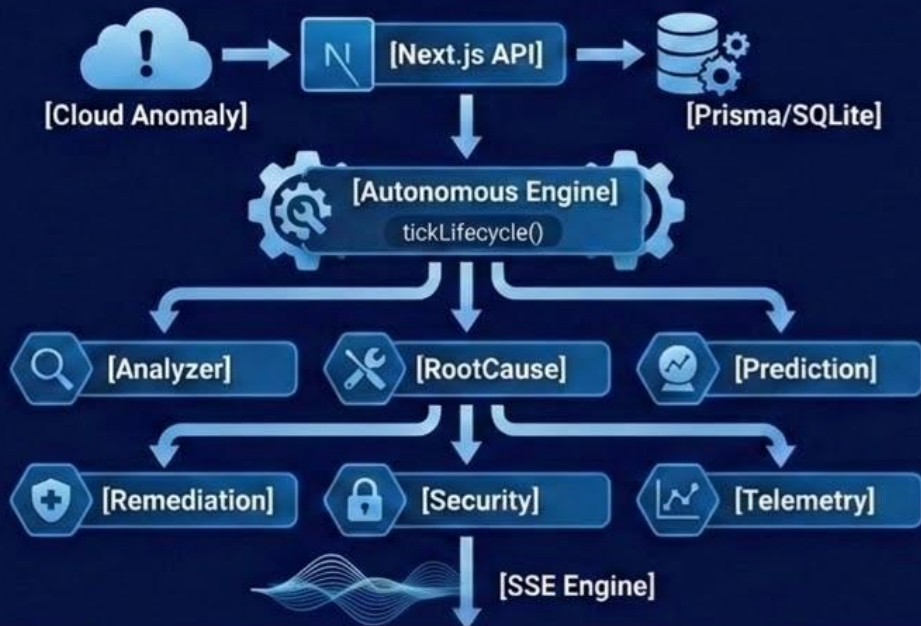


Operator Control Philosophy:

Fully autonomous systems still require human guardrails. The global and per-agent stop/resume controls ensure that operators can freeze any part of swarm with immediate backend effect — not just a UI state change.

METHODOLOGY

CRISISSWARM ARCHITECTURE



1. DETECT & ASSESS

Ingests cloud anomalies; initiates autonomous engine analysis.

2. DIAGNOSE & PREDICT

Pinpoints root causes and forecasts potential system impacts.

3. RESOLVE & SECURE

Deploys automated remediations and validates security protocols.



WORKING PROTOTYPE

 Live Demo WEB APP VIDEO – "[Zero-Human Incident Response in 50 Seconds](#)"

Demo Video explanation: [I PREFER YOU TO SEE THIS VIDEO]

https://drive.google.com/file/d/1qUtry4IfLKSZDcyYNanC7EXbql5DXQhS/view?usp=drive_link

Live App URL: [<https://calm-ocean-08f5d1400.7.azurestaticapps.net>]

GitHub Repository: [<https://github.com/Princedeepu381/CrisisSwarm>]